

Parts Manual



See "General Information"
for New Parts Manual
Features.

3406C Generator Set

C7A1-Up (Generator)
F5A1-Up (Generator)
F6B1-Up (Generator)
C8G1-Up (Generator Set)
1DZ1-Up (Engine)

This Parts Manual is also available in .PDF Format on Compact Disc
(CD-ROM)- Caterpillar Form No. SERP4245

Product and Dealer Information

Note: For product identification plate locations, see the section “Product Identification Information” in the Operation and Maintenance Manual.

Delivery Date: _____

Product Information

Model: _____

Product Identification Number: _____

Engine Serial Number: _____

Transmission Serial Number: _____

Generator Serial Number: _____

Attachment Serial Numbers: _____

Attachment Information: _____

Customer Equipment Number: _____

Dealer Equipment Number: _____

Dealer Information

Name: _____ Branch: _____

Address: _____

	<u>Dealer Contact</u>	<u>Phone Number</u>	<u>Hours</u>
Sales:	_____	_____	_____
Parts:	_____	_____	_____
Service:	_____	_____	_____

MEMORANDA

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GENERAL INFORMATION

1. ENGINE IDENTIFICATION

Caterpillar engines are identified with Serial Numbers and Arrangement Numbers. In some cases, Modification Numbers are also used. The serial number plate which shows these numbers is mounted on the engine.

Caterpillar dealers need this information to determine which components were included on the engine when it was assembled at the factory. This permits accurate identification of replacement part numbers.



New Feature

Effective First Quarter 2001 the Caterpillar Product Identification Number (PIN) changed from 8 to 17 characters. In an effort to provide uniform equipment identification, Caterpillar and other construction equipment manufacturers are moving to comply with the latest version of the product identification numbering standard. Non-road machine PINs are defined by ISO 10261. The new PIN format will apply to all Caterpillar machines and generator sets. The PIN plates and frame marking will display the 17 character PIN. The new format will look like the following:

* C A T 0 7 8 9 B G 6 S L 1 2 3 4 5 *

1 2 3 4

Where:

1. Caterpillar's World Manufacturing Code (characters 1-3)
2. Machine Descriptor (characters 4-8)
3. Check Character (character 9)
4. Machine Indicator Section (MIS) or Product Sequence Number (characters 10-17). These were previously referred to as the Serial Number.

Machines and generator sets produced before First Quarter 2001 will maintain their 8 character PIN format.

Components such as engines, transmissions, axles, etc. and work tools will continue to use an 8 character Serial Number (S/N).

2. ORDERING PARTS



WARNING

When replacement parts are required for this product Caterpillar recommends using Caterpillar replacement parts or parts with equivalent specifications including, but not limited to, physical dimensions, type, strength and material.

Failure to heed this warning can lead to premature failures, product damage, personal injury or death.

Quality Caterpillar replacement parts are available from Caterpillar dealers throughout the world. Their parts inventories are up to date and include all parts normally required to protect your investment in Caterpillar machines. Dealers may offer repair kits or remanufactured parts to allow selection of the most effective repair alternative for a particular situation. When ordering parts, your order should specify the quantity, part number, part name, and the serial number/product identification number of the engine for which the parts are needed.

3. PARTS MANUAL ORGANIZATION

Product information in this manual is presented as "information elements" that represent all of the components for the specific model. Engine Arrangement and Seat Group are examples of information elements. The information elements are organized alphabetically by part name and secondarily by part number within each major section of the manual.

4. TABLE OF CONTENTS

A table of contents (TOC) is found at the beginning of the manual. The TOC lists each section of the manual with a complete list of all information elements organized as they appear in the manual. Page numbers are provided for quick reference to detailed parts identification illustrations and serviceable consist lists.

5. MAINTENANCE PARTS INDEX

The Maintenance Parts Index, located near the beginning of the manual, references most frequently used maintenance part numbers, providing description, quantity, usage and page number. This information is organized alphabetically by part description.

GENERAL INFORMATION

6. INDEX

The index located near the front of the manual is an alphabetical listing of all information elements included in the manual. Page numbers are provided for quick reference to detailed parts identification illustrations and serviceable consist lists.

7. PART NUMBER INDEX

A numerical index listing all part numbers and the corresponding page number(s) appears at the back of the manual.

8. CAPTIONS

Captions located at the beginning of each information element identify the part number and part name along with additional descriptive information. S/N information found in each caption should be used to select correct information for a specific engine. Field replacement options, identification of optional attachment components, and where used ("part of") information is also provided.

The first line of a caption is shown in larger type font to indicate the beginning of an information element. Captions for additional pages that may be required to illustrate an information element will be shown in standard type font and will include the term (*contd.*) .

9. NON-SERVICED PARTS

In some instances it is necessary to display non-serviced parts that are a link to lower level serviceable consist lists. These non-serviced part numbers are shown in *italic* type indicating that they are not available. All consist lists have not been converted to reflect this enhancement.

10. PARTS LIST REFERENCE NUMBERS

Numbers shown in the Ref No column correspond to numbers used in the associated graphic(s). An alphabetic suffix may be added to a reference number to identify lower level consist items. Numbers shown in the Graphic Ref column refer to the graphic number identifier displayed in the lower left corner of each illustration. The Graphic Ref number may be used in combination with the item Ref No to determine the correct part number. There may be intentional omission of a Ref No in a consist list as information is updated to reflect the latest serviceable part numbers.

11. INDENTED PART NAMES

When a part name is indented in a parts list, it is part of (included in) the group or assembly under which it is indented.

12. ILLUSTRATION REFERENCE LETTERS

When necessary, illustrations contain reference letters (A, B, C etc.) that are intended to track lines and harnesses from one point to another. They are also used to show where to reconnect illustrations that have been separated.

13. ABBREVIATIONS AND SYMBOLS

O.D. – Outside Diameter
I.D. – Inside Diameter
A – Not Part Of This Group
B – Use As Required
C – Change From Previous Type
D – Order By The Meter
E – Order By The Centimeter
F – Not Shown
G – Order By The Inch
I – Refer To Hydraulic Information System
M – Metric Part
R – Remfg Part May Be Available
Y – Separate Illustration
Z – Not Serviced Separately

14. REPLACEMENT PARTS FOR GENERATORS

Only those generators which are sold and serviced exclusively by "Caterpillar" Dealers are illustrated on the following pages. Externally-regulated "Caterpillar" Diesel Electric Sets are equipped with generators made by other manufacturers. Parts for these generators are NOT serviced by Caterpillar Inc. Parts lists and parts should be ordered from the manufacturer whose name is shown on the Generator Name Plate, always giving the generator rating and serial number shown.



New Feature

15. DESCRIPTION OF TYPE CHANGES

Type - A type is defined as any configuration change that requires an additional Information Element for an Arrangement, Group, or Assembly. If serial number breaks for types are not available, a type # (Type 1, Type 2, etc.) will be displayed in the caption. These "type" changes are identified with a "C" note (change from previous type) in the Parts List. Only the "types" that apply to this Parts Manual will be included. Additional types may exist causing "C" notes to appear on an Information Element when no other types are shown.

GENERAL INFORMATION

16. <END>

This symbol indicates the end of an information element.

NOTE:

Continuing improvement and advancement of product design may cause changes to your machine which may not be included in this publication. Whenever a question arises regarding your Caterpillar product or this publication, please consult your Caterpillar dealer for the latest available information.

REPAIR ALTERNATIVES

REMANUFACTURED COMPONENTS

As an option when making repairs consider Caterpillar Remanufactured Components. Components that are available through the Caterpillar Remanufactured Program are identified three ways in the parts book:

- with the letter R in the note field of the parts list
- with an R* at the beginning of the first line of the caption
- with an *R at the end of the first line of the caption

Typical components included in the Remanufacturing Program include:

ALTERNATORS
CONNECTING RODS
CRANKSHAFTS - UNDERSIZE
CRANKSHAFTS - UPGRADE TO NEW
CYLINDER HEADS
ELECTRONIC CONTROL MODULES (ECM)
ELECTRONIC SENSORS
FUEL INJECTORS
FUEL NOZZLES
FUEL PUMPS
GOVERNORS
OIL PUMPS
PISTONS
SHORT BLOCKS
STARTERS
TURBOCHARGERS - COMPLETE
TURBOCHARGER CARTRIDGES
WATER PUMPS

Caterpillar Remanufactured engines for many engine arrangements are also available.

SUPPLEMENTAL COOLANT ADDITIVES

CUSTOMER BENEFITS:

Protects against cavitation, foam, erosion, corrosion, and scale buildup in cooling systems.

- Provides measured coolant conditioner.

DESCRIPTION:

Additives deplete from the coolant with normal operation. For this reason, Supplemental Coolant Additives (SCA) must be added to all heavy duty coolants at regular intervals. In addition, when not using a fully formulated, pre-charged antifreeze, like Caterpillar's DEAC, an initial charge of SCA must be added to the cooling system. The amount of SCA added to the system is dependent on the capacity of the system.

The SCA for initial fill is provided as a liquid. SCA's for maintenance intervals are available in liquid form and as a spin-on element. (No Spin-on available for 3600 Family of Engines)

NOTE:

The amount of SCA added at initial fill is not the same as the amount added at maintenance intervals.

The coolant additive elements can be used with any ethylene or propylene glycol type antifreeze which meets ASTM D4985 or ASTM D5345 Specifications.

NOTE:

For use with standard heavy duty coolant only, not for use with Extended Life Coolant.

PARTS NEEDED:

Cooling System Capacity		Liquid SCA at Initial Fill	Liquid SCA at Maintenance*	Spin-on SCA Maintenance Element*
Liters	Gallons			
22-30	(6-8)	3P-2044 (1)	6V-3542 (1)	111-2370 (1)
30-38	(8-10)	3P-2044 (1) 6V-3542 (1)	111-2372 (1)	111-2369 (1)
38-49	(10-13)	3P-2044 (1) 8T-1589 (1)	111-2372 (1)	111-2369 (1)
49-64	(13-17)	3P-2044 (2)	8T-1589 (1)	9N-3368 (1)
64-83	(17-22)	3P-2044 (2) 8T-1589 (1)	111-2372 (1) 6V-3542 (1)	111-2371 (1)
83-114	(22-30)	3P-2044 (3) 8T-1589 (1)	3P-2044 (1)	9N-3718 (1)
114-163	(30-43)	3P-2044 (5)	3P-2044 (1) 6V-3542 (1)	111-2371 (2)
163-242	(43-64)	3P-2044 (8)	3P-2044 (2)	9N-3718 (2)

* Normal maintenance period is 250 hrs. See your Operation and Maintenance Manual for complete coolant maintenance instructions.

() Indicates quantity required.

MAINTENANCE PARTS

DESCRIPTION	PART NUMBER	QTY	WHERE USED	PAGE
AIR INLET AND EXHAUST SYSTEM				
FILTER ELEMENT AS-AIR (PRIMARY)	142-1340	1	AIR CLEANER GP	100
	151-7737	1	AIR CLEANER GP	99
FILTER ELEMENT AS-AIR (SECONDARY)	142-1403	1	AIR CLEANER GP	98
	189-0202	1	AIR CLEANER GP	97
		1	AIR CLEANER GP	101
BASIC ENGINE				
PLUG-DRAIN (MAGNETIC) (OIL PAN)	206-7331	1	PAN GP-OIL	49
PLUG-DRAIN (OIL PAN)	9X-6451	1	PAN GP-OIL	49
V-BELT (ALTERNATOR)	4N-8216	1	MOUNTING GP-ALTERNATOR	47
V-BELT SET (INCLUDES 3-BELTS) (FAN DRIVE)	6N-6656	1	DRIVE GP-FAN	36
COOLING SYSTEM				
CAP AS-RADIATOR	153-1403	1	RADIATOR GP	87
		1	RADIATOR GP	90
HOSE (LOWER RADIATOR)	209-6964	1	RADIATOR GP	88
HOSE (RADIATOR LOWER)	209-6964	1	RADIATOR GP	90
HOSE (RADIATOR UPPER)	209-6963	1	RADIATOR GP	90
HOSE (UPPER RADIATOR)	209-6963	1	RADIATOR GP	88
REGULATOR-TEMPERATURE (92-DEG C)	248-5513	1	LINES GP-WATER	82
ELECTRICAL AND STARTING SYSTEM				
BATTERY (12-VOLT, MAINTENANCE FREE, WET)	9X-3404	2	BATTERY GP	190
BATTERY GP (12-VOLT) (DRY)	9X-9730	2	BATTERY GP	189
CIRCUIT BREAKER AS (1250-AMPERE)	227-2788	1	CIRCUIT BREAKER GP	199
CIRCUIT BREAKER AS (20-AMPERE)	200-8577	1	BOX GP-CONNECTION	191
CIRCUIT BREAKER AS (630-AMPERE)	237-2612	1	CIRCUIT BREAKER GP	197
CIRCUIT BREAKER AS (800-AMPERE)	227-2794	1	CIRCUIT BREAKER GP	198
CIRCUIT BREAKER GP (60-AMPERE)	6T-3645	1	BOX GP-CONNECTION	191
FUSE (1-AMPERE) (US SOURCED)	8K-4644	10	MODULE GP-INTERFACE	236
FUSE (10-AMPERE) (EUROPEAN SOURCED)	143-5080	1	BATTERY CHARGER GP	186
		1	BOX GP-CONNECTION	191
		1	CONTACTOR GP	210
		1	CONTACTOR GP	211
		1	CONTACTOR GP	216
		1	CONTACTOR GP	218
		1	FUSE GP	226
FUSE (10-AMPERE) (US SOURCED)	6V-7802	1	BATTERY CHARGER GP	186
		1	BOX GP-CONNECTION	191
		1	CONTACTOR GP	210
		1	CONTACTOR GP	211
		1	CONTACTOR GP	216
		1	CONTACTOR GP	218
		1	FUSE GP	226
FUSE (15-AMPERE) (US SOURCED)	9U-6398	1	FUSE GP	225
		1	WIRING GP-AIR SHUTOFF	305
FUSE (16-AMPERE) (EUROPEAN SOURCED)	151-7710	1	FUSE GP	225

MAINTENANCE PARTS

DESCRIPTION	PART NUMBER	QTY	WHERE USED	PAGE
FUSE (2-AMPERE)	143-4471 7W-0482	1	WIRING GP-AIR SHUTOFF	305
		1	CONTACTOR GP	220
		1	CONTACTOR GP	220
FUSE (2-AMPERE) (EUROPEAN SOURCED)	143-4471	3	WIRING GP-PANEL	312
		1	BATTERY CHARGER GP	187
		1	CONTACTOR GP	206
		1	CONTACTOR GP	208
		1	CONTACTOR GP	209
		1	CONTACTOR GP	210
		1	CONTACTOR GP	211
		1	CONTACTOR GP	212
		1	CONTACTOR GP	214
		1	CONTACTOR GP	216
FUSE (2-AMPERE) (US SOURCED)	7W-0482	1	CONTACTOR GP	218
		2	WIRING GP-PANEL	311
		1	BATTERY CHARGER GP	187
		1	CONTACTOR GP	206
		1	CONTACTOR GP	208
		1	CONTACTOR GP	209
		1	CONTACTOR GP	210
		1	CONTACTOR GP	211
		1	CONTACTOR GP	212
		1	CONTACTOR GP	214
FUSE (4-AMPERE)	190-2557 238-3255	1	CONTACTOR GP	216
		1	CONTACTOR GP	218
		2	WIRING GP-PANEL	311
FUSE (4-AMPERE) (EUROPEAN SOURCED)	238-3255	1	CONTACTOR GP	220
		1	CONTACTOR GP	220
		1	BATTERY CHARGER GP	187
FUSE (4-AMPERE) (US SOURCED)	190-2557	1	CONTACTOR GP	206
		1	CONTACTOR GP	212
		1	CONTACTOR GP	214
		1	CONTACTOR GP	216
		1	CONTACTOR GP	218
		1	BATTERY CHARGER GP	187
		1	CONTACTOR GP	206
		1	CONTACTOR GP	212
		1	CONTACTOR GP	214
		1	CONTACTOR GP	216
FUSE (6-AMPERE) (EUROPEAN SOURCED)	143-4472	1	CONTACTOR GP	218
		2	BATTERY CHARGER GP	185
		1	BATTERY CHARGER GP	186
		2	BATTERY CHARGER GP	187
		2	CONTACTOR GP	208
		2	CONTACTOR GP	209
		1	CONTACTOR GP	210
		1	CONTACTOR GP	211
		2	CONTACTOR GP	212
		2	CONTACTOR GP	214
		1	CONTACTOR GP	216
		1	CONTACTOR GP	218

MAINTENANCE PARTS

DESCRIPTION	PART NUMBER	QTY	WHERE USED	PAGE
FUSE (6-AMPERE) (US SOURCED)	190 - 2558	2	BATTERY CHARGER GP	185
		1	BATTERY CHARGER GP	186
		2	BATTERY CHARGER GP	187
		1	CONTACTOR GP	208
		1	CONTACTOR GP	209
		1	CONTACTOR GP	210
		1	CONTACTOR GP	211
		2	CONTACTOR GP	212
		2	CONTACTOR GP	214
		1	CONTACTOR GP	216
		1	CONTACTOR GP	218
KEY (BATTERY DISCONNECT)	8H - 5306	1	SWITCH GP-DISCONNECT	281
SHUNT TRIP AS	223 - 4234	1	SHUNT-TRIP	262
FUEL SYSTEM				
CAP AS-FUEL FILLER (FUEL TANK)	143 - 0853	1	BASE GP-FUEL TANK	128
ELEMENT-WATER SEPARATOR	133 - 5673	1	SEPARATOR GP-WATER	177
ELEMENT-WATER SEPARATOR (PRIMARY)	133 - 5673	1	FILTER GP-WATER SEP & FUEL	145
FILTER ELEMENT-FUEL (PRIMARY)	9M - 2342	1	FILTER GP-FUEL	143
FILTER-FUEL (ADVANCED EFFICIENCY)	1R - 0749	1	FILTER GP-FUEL	142
GASKET (FUEL PRIMING PUMP)	1P - 0436	1	PUMP GP-FUEL PRIMING	170
SCREEN AS (FUEL INJECTION PUMP)	8N - 6680	1	PUMP GP-GOV & FUEL INJ	175
GENERATORS				
FUSE (10-AMPERE)	6V - 7802	3	REGULATOR GP-VOLTAGE	436
		3	REGULATOR GP-VOLTAGE	438
		3	REGULATOR GP-VOLTAGE	439
FUSE (10-AMPERE) (US SOURCED)	6V - 7802	3	REGULATOR GP-VOLTAGE	435
LUBRICATION SYSTEM				
BREATHING AS (CRANKCASE)	9Y - 2988	1	BREATHING GP	60
CAP AS-FILLER (ENGINE OIL)	6N - 2985	1	FILLER GP-ENGINE OIL	62
FILTER AS-ENGINE OIL (STANDARD EFFICIENCY)	1R - 0716	1	FILTER GP-ENGINE OIL	63

MAINTENANCE PARTS

DESCRIPTION	PART NUMBER	QTY	WHERE USED	PAGE
OPERATOR STATION				
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	315		
WIRING GP-PANEL—GENERATOR RUN	316		
WIRING GP-PANEL—GOVERNOR	310		
WIRING GP-PANEL—			
MOTORISED GOVERNOR	314		
WIRING GP-PANEL—SYNCHRONIZER	311		

MEMORANDA

ENGINE ARRANGEMENT

150-2981 ENGINE AR

S/N: 1DZ6500-UP

50 HERTZ-3 PERCENT GOVERNOR (1500 RPM)

PART OF 293-0919 GENERAL AR

FOR USE WITH SAE NO. 1 FLYWHEEL HOUSING
AN ATTACHMENT

SMCS-1000

i03397848

NOTE	PART NUMBER	QTY	PART NAME						SEE PAGE
			1	2	3	4	5	6 (PRODUCT LEVEL)	
Y	1W-8347	1	AFTERCOOLER GP-WATER						94
YR	5N-5692	1	ALTERNATOR GP-CHARGING						182
Y	2N-8220	1	BREATHER GP						60
Y	7W-3798	1	CAMSHAFT GP						7
Y	1W-7411	1	CONTROL GP-GOVERNOR						131
Y	7W-2206	1	COOLER GP-ENGINE OIL						76
Y	1W-5107	1	COVER GP-CYLINDER BLOCK						10
Y	4N-4792	1	COVER GP-FLYWHEEL HOUSING						12
Y	1W-1395	1	COVER GP-FRONT HOUSING						13
			S/N: 1DZ6500-18750						
Y	219-1407	1	COVER GP-FRONT HOUSING						15
			S/N: 1DZ18751-UP						
Y	4W-1727	1	COVER GP-FUEL TRANSFER						136
Y	4N-4953	1	COVER GP-VALVE MECHANISM						16
Y	4N-0741	1	CRANKSHAFT GP						17
Y	152-7642	1	CYLINDER BLOCK GP						26
YR	4W-2436	1	CYLINDER HEAD GP						29
Y	4P-5894	1	DAMPER & PULLEY GP						34
Y	8N-9653	1	DRIVE GP-ACCESSORY						35
Y	7N-8840	1	DRIVE GP-FAN						36
Y	4P-1365	1	DRIVE GP-GOVERNOR & FUEL PUMP						137
Y	6L-2927	1	ELBOW GP-EXHAUST						106
Y	4N-6006	1	FASTENER GP-FRONT HOUSING						38
Y	1W-7376	1	FASTENER GP-FUEL INJ PUMP						140
Y	4N-2481	1	FASTENER GP-OIL PUMP MOUNTING						61
Y	6N-1016	1	FILLER GP-ENGINE OIL						62
Y	154-4326	1	FILTER GP-ENGINE OIL						63
Y	4N-1981	1	FILTER GP-FUEL						142
Y	147-3081	1	FILTER GP-FUEL						144
Y	7N-3060	1	FLYWHEEL GP						39
Y	6N-8878	1	GAUGE GP-FUEL PRESSURE						486
Y	7N-8084	1	GAUGE GP-OIL LEVEL (DIPSTICK) (ENGINE)						65
Y	8N-9246	1	GEAR GP-FRONT						40
Y	106-7410	1	HOUSING GP-FLYWHEEL						43
Y	1W-1468	1	HOUSING GP-FRONT						44
Y	4N-1331	1	LIFTING GP-ENGINE						46
Y	4N-2474	1	LINES GP-AFTERCOOLER WATER						79
Y	6N-5015	1	LINES GP-AIR						110
Y	4N-1128	1	LINES GP-ENGINE OIL						67
Y	141-4149	1	LINES GP-ENGINE OIL DRAIN						69
Y	2W-3452	1	LINES GP-FUEL FILTER						153
Y	9Y-3070	1	LINES GP-FUEL INJECTION						158
Y	1W-4203	1	LINES GP-TURBOCHARGER OIL						71
Y	4N-1641	1	LINES GP-WATER						82
Y	4N-0754	1	MANIFOLD GP-EXHAUST						112
Y	7W-1340	1	MOUNTING GP-ALTERNATOR						47
Y	4N-0756	1	PAN GP-OIL (ENGINE)						49
YR	7E-0491	6	PISTON & ROD GP						50
Y	2W-4381	1	PUMP & MTG GP-FUEL TRANSFER						164
YR	161-4113	1	PUMP GP-ENGINE OIL						73
Y	145-8855	1	PUMP GP-FUEL PRIMING						170
Y	2W-8543	1	PUMP GP-GOV & FUEL INJ						174
Y	9N-3288	1	PUMP GP-WATER						84

ENGINE ARRANGEMENT

150-2981 ENGINE AR (contd.)

i03397848

NOTE	PART NUMBER	QTY	PART NAME						SEE PAGE
			1	2	3	4	5	6 (PRODUCT LEVEL)	
Y	4P-7610	1	SENSOR GP-SPEED						260
Y	9L-6671	1	SHIELD GP-MANIFOLD						122
Y	2W-8198	1	SHUTOFF GP-ELECTRICAL						263
Y	2W-9161	1	SHUTOFF GP-MANUAL						179
Y	4N-1062	1	STARTING MOTOR GP-ELECTRIC						270
Y	9Y-1856	1	SUPPORT GP-ENGINE						55
Y	1W-4209	1	TURBOCHARGER GP						125
Y	242-1369	1	VALVE GP-COOLANT SAMPLING						93
Y	4W-2461	1	VALVE MECHANISM GP						56

PLEASE USE THE GASKET KIT LINK IN SIS OR
CONTACT YOUR LOCAL DEALER FOR THE LATEST
GASKET KIT INFORMATION

R-REMFG PART MAY BE AVAILABLE
Y-SEPARATE ILLUSTRATION

ENGINE ARRANGEMENT

263-2539 ENGINE AR

S/N: 1DZ1-UP

PART OF 263-2538 GENERAL AR

SMCS-1000

i03297120

NOTE	PART NUMBER	QTY	PART NAME						SEE PAGE
			1	2	3	4	5	6 (PRODUCT LEVEL)	
Y	1W-8347	1	AFTERCOOLER GP-WATER						94
YR	5N-5692	1	ALTERNATOR GP-CHARGING						182
			S/N: 1DZ10334-UP						
Y	2N-8220	1	BREATHING GP						60
Y	7W-3798	1	CAMSHAFT GP						7
Y	1W-7411	1	CONTROL GP-GOVERNOR						131
Y	7W-2206	1	COOLER GP-ENGINE OIL						76
Y	1W-5107	1	COVER GP-CYLINDER BLOCK						10
Y	4N-4792	1	COVER GP-FLYWHEEL HOUSING						12
Y	1W-1395	1	COVER GP-FRONT HOUSING						13
			S/N: 1DZ1-17807						
Y	219-1407	1	COVER GP-FRONT HOUSING						15
			S/N: 1DZ17808-UP						
Y	4W-1727	1	COVER GP-FUEL TRANSFER						136
Y	4N-4953	1	COVER GP-VALVE MECHANISM						16
Y	4N-0741	1	CRANKSHAFT GP						17
Y	122-9276	1	CYLINDER BLOCK GP						24
YR	110-5095	1	CYLINDER HEAD GP						31
Y	4P-5894	1	DAMPER & PULLEY GP						34
Y	8N-9653	1	DRIVE GP-ACCESSORY						35
Y	7N-8840	1	DRIVE GP-FAN						36
Y	110-1282	1	DRIVE GP-GOVERNOR & FUEL PUMP						139
Y	6L-2927	1	ELBOW GP-EXHAUST						106
Y	4N-6006	1	FASTENER GP-FRONT HOUSING						38
Y	1W-7376	1	FASTENER GP-FUEL INJ PUMP						140
Y	4N-2481	1	FASTENER GP-OIL PUMP MOUNTING						61
Y	6N-1016	1	FILLER GP-ENGINE OIL						62
Y	154-4326	1	FILTER GP-ENGINE OIL						63
Y	4N-1981	1	FILTER GP-FUEL (SECONDARY)						142
Y	147-3081	1	FILTER GP-FUEL (PRIMARY)						144
Y	7N-3060	1	FLYWHEEL GP						39
Y	6N-8878	1	GAUGE GP-FUEL PRESSURE						486
Y	7N-8084	1	GAUGE GP-OIL LEVEL (DIPSTICK) (ENGINE)						65
Y	8N-9246	1	GEAR GP-FRONT						40
Y	106-7410	1	HOUSING GP-FLYWHEEL						43
Y	1W-1468	1	HOUSING GP-FRONT						44
Y	4N-1331	1	LIFTING GP-ENGINE						46
Y	4N-2474	1	LINES GP-AFTERCOOLER WATER						79
Y	8N-4398	1	LINES GP-AIR						111
Y	4N-1128	1	LINES GP-ENGINE OIL						67
Y	141-4149	1	LINES GP-ENGINE OIL DRAIN						69
Y	2W-3452	1	LINES GP-FUEL FILTER						153
Y	131-0188	1	LINES GP-FUEL INJECTION						160
Y	4N-5394	1	LINES GP-TURBOCHARGER OIL						72
Y	4N-1641	1	LINES GP-WATER						82
Y	4W-4319	1	MANIFOLD GP-EXHAUST						113
Y	7W-1340	1	MOUNTING GP-ALTERNATOR						47
			S/N: 1DZ10334-UP						
Y	4N-0756	1	PAN GP-OIL						49
Y	130-4614	6	PISTON & ROD GP						51
Y	2W-4381	1	PUMP & MTG GP-FUEL TRANSFER						164
YR	161-4113	1	PUMP GP-ENGINE OIL						73
Y	145-8855	1	PUMP GP-FUEL PRIMING						170
Y	131-1955	1	PUMP GP-GOV & FUEL INJ						175
Y	9N-3288	1	PUMP GP-WATER						84
Y	4P-7610	1	SENSOR GP-SPEED (ENGINE)						260

ENGINE ARRANGEMENT

263-2539 ENGINE AR (contd.)

i03297120

NOTE	PART NUMBER	QTY	PART NAME						SEE PAGE
			1	2	3	4	5	6 (PRODUCT LEVEL)	
Y	9L-6671	1	SHIELD GP-MANIFOLD						122
Y	2W-8198	1	SHUTOFF GP-ELECTRICAL						263
Y	2W-9161	1	SHUTOFF GP-MANUAL						179
Y	4N-1062	1	STARTING MOTOR GP-ELECTRIC						270
Y	9Y-1856	1	SUPPORT GP-ENGINE						55
Y	130-5622	1	TURBOCHARGER GP						127
Y	242-1369	1	VALVE GP-COOLANT SAMPLING						93
Y	4W-2461	1	VALVE MECHANISM GP						56

PLEASE USE THE GASKET KIT LINK IN SIS OR
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GASKET KIT INFORMATION

R-REMFG PART MAY BE AVAILABLE
Y-SEPARATE ILLUSTRATION

ENGINE ARRANGEMENT

263-2538 GENERAL AR

S/N: C8G1-UP

50 HERTZ - 3 % GOVERNOR, 1500 RPM
FOR USE WITH SAE NO. 1 FLYWHEEL HOUSING

SMCS-1000, 4450

i02193958

NOTE	PART NUMBER	QTY	PART NAME						SEE PAGE
			1	2	3	4	5	6 (PRODUCT LEVEL)	
Y	263-2539	1	ENGINE	AR					3
Y-SEPARATE ILLUSTRATION									

ENGINE ARRANGEMENT

293-0919 GENERAL AR

S/N: C8G1-UP

50 HERTZ, 3 PERCENT GOVERNOR, 1500 RPM
FOR USE WITH SAE NO. 1 FLYWHEEL HOUSING
AN ATTACHMENT

SMCS-1000, 4450

i02575772

NOTE	PART NUMBER	QTY	PART NAME						SEE PAGE
			1	2	3	4	5	6 (PRODUCT LEVEL)	
Y	150-2981	1	ENGINE AR						1
Y-SEPARATE ILLUSTRATION									

BASIC ENGINE

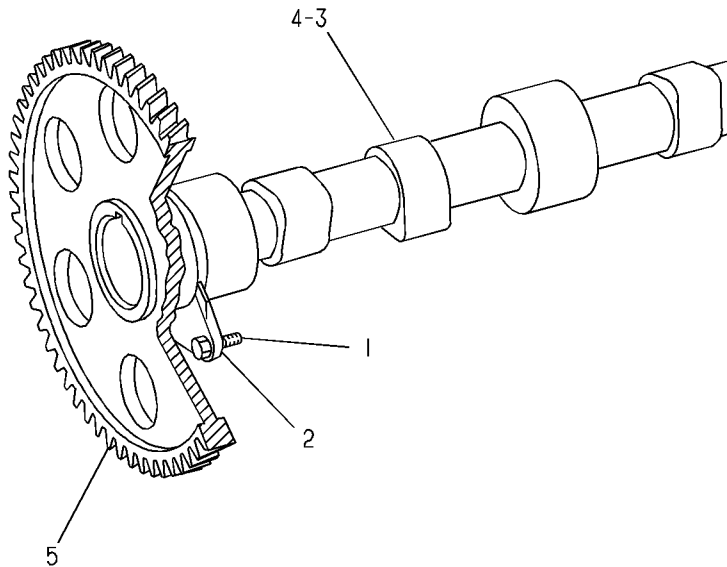
7W-3798 CAMSHAFT GP

S/N: 1DZ1-UP
ALSO AN ATTACHMENT

SMCS-1210

i01628739

NOTE	REF NO	GRAPHIC REF	PART NUMBER	QTY	PART NAME						SEE PAGE
					1	2	3	4	5	6 (PRODUCT LEVEL)	
	1	1	0S-0509	2	BOLT (3/8-16X0.75-IN)						
	2	1	4N-0342	1	PLATE						
R	3	1	100-7408	1	CAMSHAFT AS						
	4	1	1B-8705	1	KEY-WOODRUFF						
	5	1	8N-8254	1	GEAR-CAMSHAFT (64-TEETH)						
R-REMFG PART MAY BE AVAILABLE											



GRAPHIC #1

<END>

g00108229

BASIC ENGINE

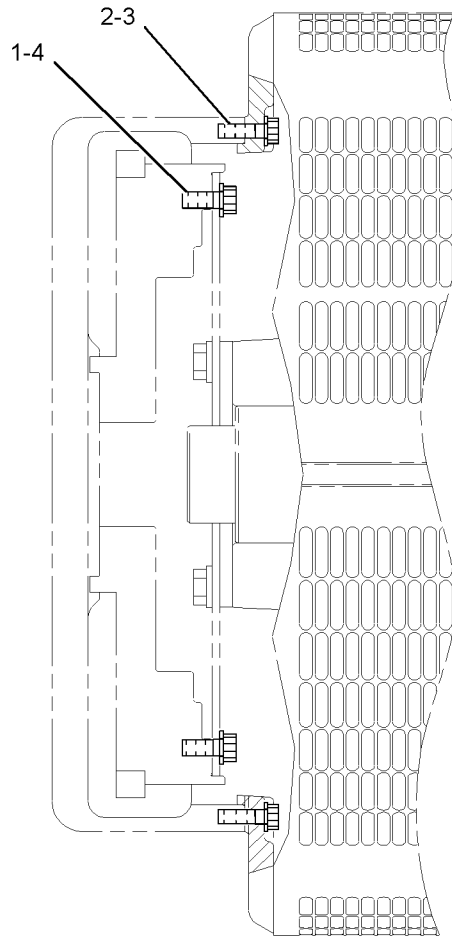
264-6040 COUPLING GP

S/N: C8G1-UP
AN ATTACHMENT

SMCS-3279

i03515901

NOTE	REF NO	GRAPHIC REF	PART NUMBER	QTY	PART NAME						SEE PAGE
					1	2	3	4	5	6 (PRODUCT LEVEL)	
	1	1	0S-1595	8	BOLT (1/2-13X1.25-IN)						
	2	1	1D-4707	12	BOLT (7/16-14X2.25-IN)						
	3	1	5P-8244	12	WASHER-HARD (12X22.5X2.5-MM THK)						
	4	1	5P-8245	8	WASHER-HARD (13.5X25.5X3-MM THK)						



GRAPHIC #1

<END>

g01024028

BASIC ENGINE

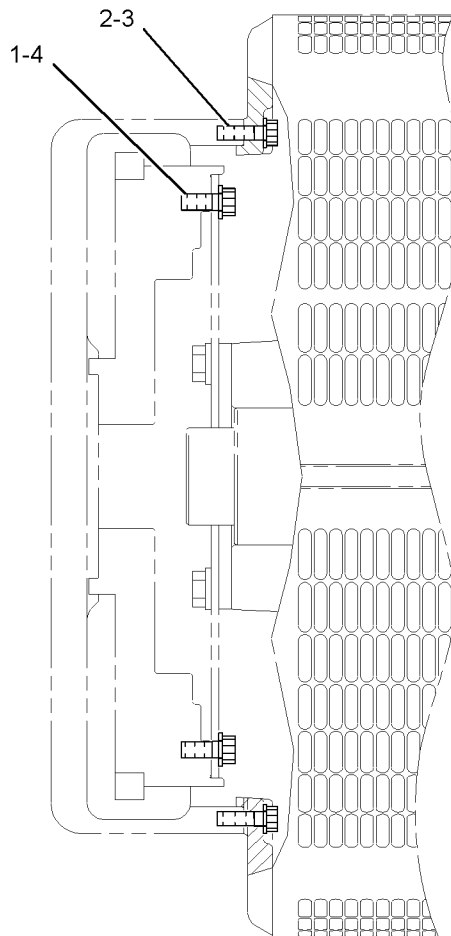
233-9351 COUPLING GP-GENERATOR

S/N: C8G1-UP
AN ATTACHMENT

SMCS-4456

i02697705

NOTE	REF NO	GRAPHIC REF	PART NUMBER	QTY	PART NAME						SEE PAGE
					1	2	3	4	5	6 (PRODUCT LEVEL)	
	1	1	0S-1595	8	BOLT (1/2-13X1.25-IN)						
	2	1	0T-0772	12	BOLT (7/16-14X1.5-IN)						
	3	1	5P-8244	12	WASHER-HARD (12X22.5X2.5-MM THK)						
	4	1	5P-8245	8	WASHER-HARD (13.5X25.5X3-MM THK)						



GRAPHIC #1

<END>

g01024028

BASIC ENGINE

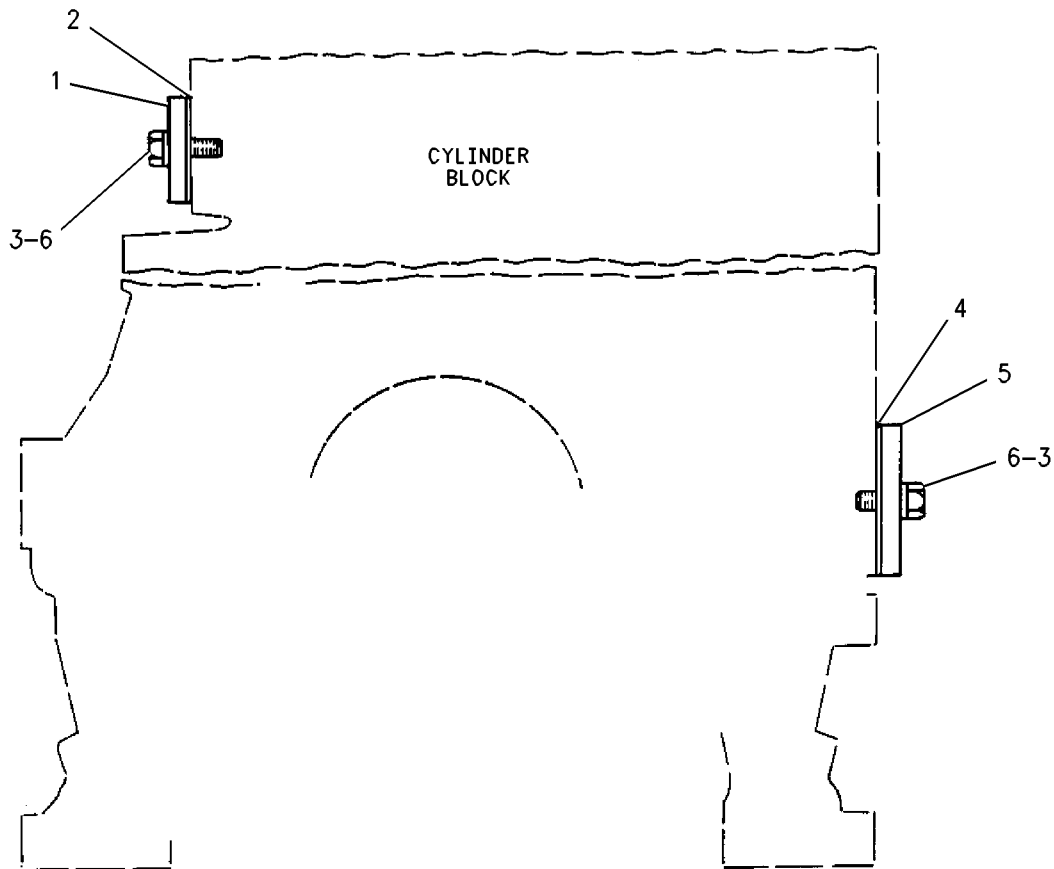
1W-5107 COVER GP-CYLINDER BLOCK

S/N: 1DZ1-UP
ALSO AN ATTACHMENT

SMCS-1201

i01628707

NOTE	REF NO	GRAPHIC REF	PART NUMBER	QTY	PART NAME						SEE PAGE
					1	2	3	4	5	6 (PRODUCT LEVEL)	
	1	1	4W-9890	1	COVER						
	2	1	4N-0699	1	GASKET						
	3	1	0S-1594	4	BOLT (3/8-16X1 IN)						
	4	1	7C-0307	1	GASKET						
	5	1	4N-0794	1	COVER						
	6	1	5M-2894	4	WASHER-HARD (10.2X18.5X2.5MM THK)						



GRAPHIC #1

<END>

g00844277

BASIC ENGINE

144-7184 COVER GP-ENGINE LIFTING EYE

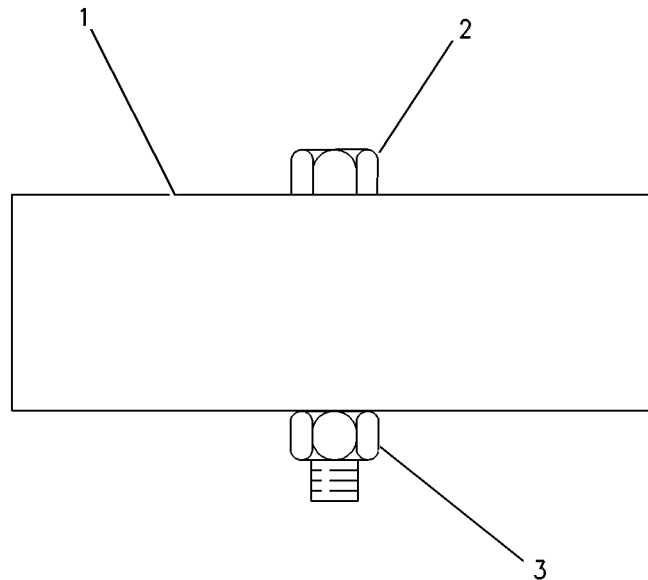
S/N: C8G1-UP

ATTACHES TO ENGINE LIFTING EYES WHEN OTHER METHODS OF LIFTING PACKAGE IS
REQUIRED
AN ATTACHMENT

SMCS-1122

i01629890

NOTE	REF NO	GRAPHIC REF	PART NUMBER	QTY	PART NAME	SEE PAGE
					1 2 3 4 5 6 (PRODUCT LEVEL)	
	1	1	144-7185	1	COVER-LIFTING EYE	
M	2	1	8T-6466	1	BOLT (M10X1.5X60-MM)	
M	3	1	8T-0389	1	LOCKNUT (M10X1.5-THD)	
M-METRIC PART						



GRAPHIC #1

<END>

g00278337

BASIC ENGINE

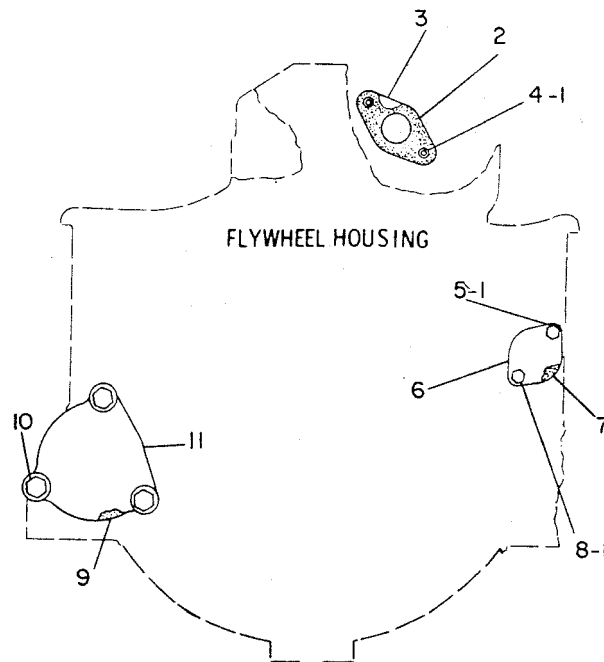
4N-4792 COVER GP-FLYWHEEL HOUSING

S/N: 1DZ1-UP
ALSO AN ATTACHMENT

SMCS-1157

i03506446

NOTE	REF NO	GRAPHIC REF	PART NUMBER	QTY	PART NAME	SEE PAGE
					1 2 3 4 5 6 (PRODUCT LEVEL)	
	1	1	5M-2894	4	WASHER-HARD (10.2X18.5X2.5-MM THK)	
	2	1	5H-0244	1	GASKET	
	3	1	8F-3030	1	FLANGE	
	4	1	0S-1588	2	BOLT (3/8-16X1.25-IN)	
	5	1	0S-1619	1	BOLT (3/8-16X3-IN)	
	6	1	2P-2772	1	COVER	
	7	1	234-8937	1	SEAL-INTEGRAL	
	8	1	0S-1594	1	BOLT (3/8-16X1-IN)	
	9	1	9Y-6089	1	GASKET	
	10	1	4B-4043	3	BOLT (5/8-11X1.375-IN)	
		1	5P-8247	3	WASHER-HARD (17X32X3.5-MM THK)	
	11	1	3H-0761	1	COVER-OPENING	



GRAPHIC #1

<END>

g00110887

BASIC ENGINE

1W-1395 COVER GP-FRONT HOUSING

S/N: 1DZ1-UP
ALSO AN ATTACHMENT

SMCS-1151, 1166

i03465541

NOTE	REF NO	GRAPHIC REF	PART NUMBER	QTY	PART NAME 1 2 3 4 5 6 (PRODUCT LEVEL)	SEE PAGE
	1	1	7W-5733	1	GASKET	
	2	1	4N-0273	1	COVER	
	3	1	5M-2894	22	WASHER-HARD (10.2X18.5X2.5-MM THK)	
	4	1	0S-1588	5	BOLT (3/8-16X1.25-IN) (MARKED X)	
	5	1	1A-5964	7	BOLT (3/8-16X4.5-IN)	
B	6	1	315-5008	2	STUD-TAPERLOCK	
B	7	1	2M-0849	2	WASHER-HARD (13.5X25.5X5-MM THK)	
	8	1	1F-7958	5	NUT-FULL (1/2-13-THD)	
C	9	1	1H-7339	1	SEAL-O-RING	
C	10	1	4P-1650	1	COVER	
C	11	1	9S-8752	10	NUT-FULL (3/8-16-THD)	
	12	1	6V-4259	2	BOLT-SHALLOW HEAD (3/8-16X0.5-IN)	
	13	1	7W-9467	1	COVER	
	14	1	1W-2488	1	COVER	
	15	1	2W-0459	1	GASKET	
	16	1	2B-2006	1	BOLT (1/2-13X4.75-IN)	
	17	1	9S-1366	1	BOLT (1/2-13X2.5-IN)	
	18	1	100-7432	1	GASKET	
	19	1	5P-8245	5	WASHER-HARD (13.5X25.5X3-MM THK)	
	20	1	5M-6214	1	PLUG-PIPE	

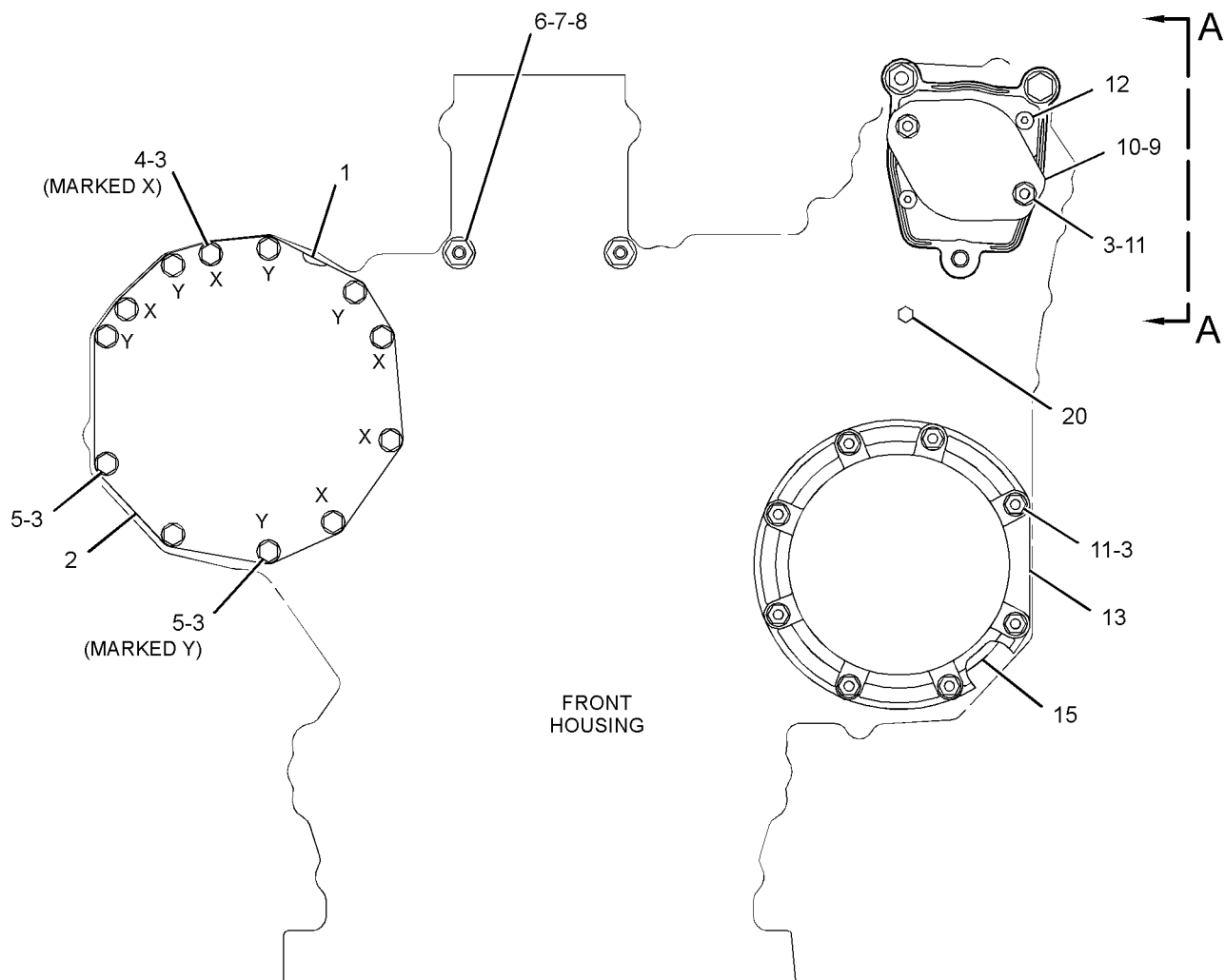
B-USE AS REQUIRED

C-CHANGE FROM PREVIOUS TYPE

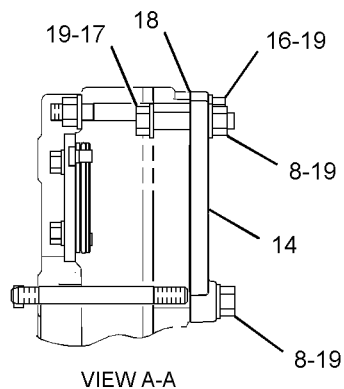
BASIC ENGINE

1W-1395 COVER GP-FRONT HOUSING (contd.)

i03465541



FRONT VIEW OF ENGINE



VIEW A-A

<END>

GRAPHIC #1

g01817153